

□ 18 Statistical Inference – The t test continued

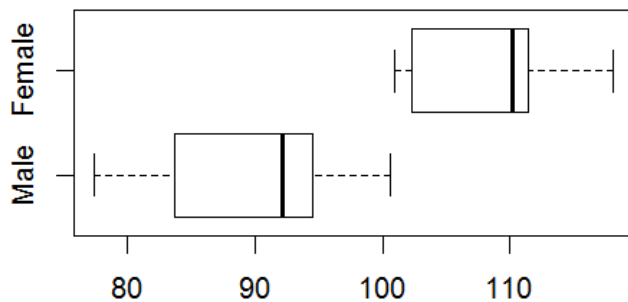
Consider example 2 again A survey is commissioned of first year students at WIT to determine if there is a difference between male and female students in terms of what average weekly rent they pay.

Suppose the following data is obtained for samples of male and female students.

Male	92.18	94.52	83.71	80.45	98.62	93.58	83.75	87.57	94.5	100.65	89.5	95.56	77.45
Female	108.52	117.12	100.94	102.25	111.42	110.13	101.09	111.03	118.03				

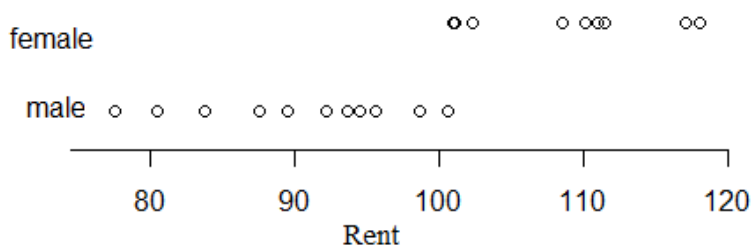
Assuming the data for males and females are stored in variables `male` and `female` respectively.

```
> boxplot(male, female, horizontal = TRUE, names=c("Male", "Female"))
```



We can see a clear indication in the boxplot that males tend to pay less than females.

Sometimes, especially if the sample sizes are small, a dot plot might be preferred.



```
> t.test(male, female)
```

Welch Two Sample t-test

data: male and female

t = -6.4368, df = 18.468, p-value = 4.123e-06

alternative hypothesis: true difference in means is not equal to 0

95 percent confidence interval:

-24.91290 -12.66881

sample estimates:

mean of x mean of y

90.15692 108.94778

The confidence interval-estimate for the difference between average rent paid for males and females is given by [-24.91, -12.67]

We can say (with 95% confidence) that males pay somewhere between €12.67 and €24.91 less than females.

As the interval excludes zero, we can say (subject to checks that we will return to later) that there is a statistically significant difference between males and females in the amount of rent they pay.

Equivalently, we say that we reject the null hypothesis that mean rent is the same for males and females at the 5% level of significance (as the confidence interval level is 95%)

Why plot the data?

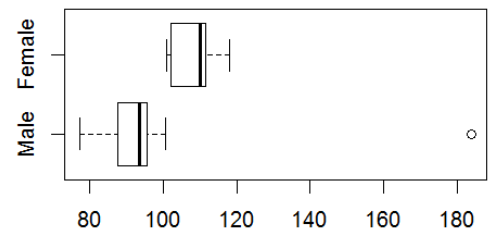
The call to `t.test` to compare the two samples can be made without generating any kind of plot. So why do we do so?

Suppose an error was made, either when the data was collected or when it was entered in to R, For example, the third male in the sample pays a weekly rent of €83.71 but suppose this value was erroneously recorded as €183.71

We can simulate such a typo with `> male[3] <- 183.71`

The boxplot now is shown opposite and the extreme value (outlier) shown.

The effect of this outlier is to raise the mean rent for males resulting in the difference between the two groups no longer being statistically significant.



Were the outlier a little bigger, it might even have led to the decision that males pay *more* rent than females.

Plotting the data will allow us to readily see possible outliers, or other issues, with the data.

Example 3 (again)

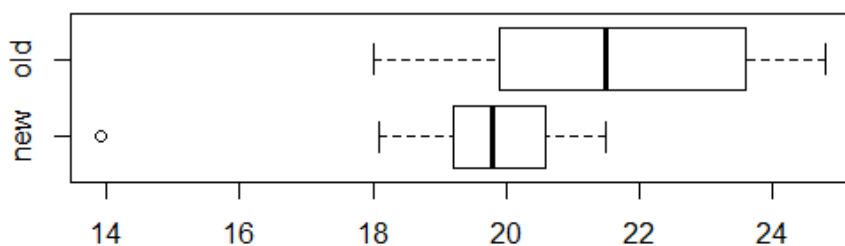
Is the mean recovery time for patients treated with a new drug shorter than those treated with an existing drug? The following data was collected for the study.

New drug	18.1	19.6	19.7	20.8	19.9	13.9	20.2	20.6	21.5	19.2
Old drug	22.6	24.8	19.9	22.8	18	20.4	23.8	20.4	23.6	19

Let Time_{old} be the recovery time for the old drug and similarly define Time_{new} .

$H_0: \text{Time}_{\text{old}} = \text{Time}_{\text{new}}$ $H_A: \text{Time}_{\text{old}} \neq \text{Time}_{\text{new}}$

A box plot is shown below.



It suggests that the new drug may lead to shorter recovery times.

But we need to be cognisant of the outlier for the new drug. Even if this is not an error, it would not be satisfactory to have a statistically significant finding (if we do) that hinges on a single observation.

```
t.test(old, new)
```

```
Welch Two Sample t-test
```

```
data: old and new
t = 2.2078, df = 17.911, p-value = 0.04054
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 0.1048161 4.2551839
sample estimates:
mean of x mean of y
 21.53      19.35
```

The confidence interval estimate (above) suggests that the new drug gives statistically significantly shorter recovery times than the old drug with an interval of [0.1, 4.3] days or 2.2 ± 2.1 days

Assuming that the various checks that we will revisit later are satisfactory, would the advice be that the new drug be adopted?

Not necessarily. There are a few reasons why we might not – in spite of the statistically significant finding.

- Small sample

It is something of a truism in statistics that more (data) is better. Not only is it more likely that a statistically significant finding will be established with a larger sample compared to a smaller sample, such a finding will be more convincing.

- Outlier

There is an outlier in the data. In a small sample, even one outlier might distort the findings. And even without the outlier, it is clearly unsatisfactory to be establishing a statistical finding that depends on one, or even a small number of observations.

- Statistical significance v practical importance.

The effect size here is 2.2 days. Patients on the new drug take on average, 19.35 days to recover whereas patients on the old drug take 21.53 days.

It is quite plausible that such a difference, while statistical significance is not practically important.

N.B. A common mistake in statistical inference is to assume that a statistically significant finding is an important finding. Loosely speaking, statistically significant means unlikely to be explained by random variation.

- The significance level is 5%

Remember that if we reject the null at the 5% level there is a 5% chance that we are making a type I error. In the context of this problem that would mean that there is a 5% chance that we will decide that there is a difference between the drugs when in fact they are the same.

The old drug, currently in use will be better understood (side effects etc) than a new one so we may wish to be a surer than the new drug is better before we elect to use it.

Consequently, we might wish to use confidence level higher than 95% - say 99%.

```
> t.test(old, new, conf.level = 0.99)
```

It is left as an exercise to show that this leads to a 99% confidence interval estimate of [-0.7, 5.0]

As this interval contains zero, we would not establish statistical significance at the 1% level – as we did at the 5% level.

Example 4

A gardener uses an expensive crop-growth promoter to enhance the yield of a fruit crop. She sets up an experiment involving 20 plants – 10 are randomly assigned to receive the growth enhancer (treatment group) throughout the growing season, the rest do not (control group). The response variable is the total yield in grams from each plant. The data is shown below.

treatment	288.3	344.5	331.8	293.8	318.4	314.7	261.4	337.2	281.3	248.1
control	300.0	266.2	315.4	298.7	264.6	229.1	306.9	269.8	260.3	267.7

Let μ_T and μ_C be the mean yield for the treatment and control groups respectively.

Form the hypothesis $H_0: \mu_T = \mu_C$ $H_A: \mu_T \neq \mu_C$

It is left as an exercise in R to show that a 95% confidence interval for $\mu_T - \mu_C$ is given by $[-4, 52]$ grams. As this interval contains zero, we do not reject H_0 .

But consider the alternative hypothesis $H_A: \mu_T \neq \mu_C$

It specifies that the two means are different. It allows that the treatment group mean may be higher than the control group but also that control group mean may be higher than the treatment group!

The gardener might argue that the latter possibility is implausible. I.e. the promoter may enhance growth, or it may have no effect. But it is implausible that it would inhibit growth.

Thus, she might elect for the alternative hypothesis $H_A: \mu_T > \mu_C$

Such an alternative hypothesis (using $>$ or $<$) is called a one-sided alternative. An alternative that uses $\neq$ is called a two-sided alternative.

Correspondingly, we can carry out a one or two-sided t test

```
t.test(treatment , control, alternative ="greater")
```

R will report a 95 percent confidence interval of $[0.95, \infty]$

This interval does not contain zero so with a one-sided test, statistical significance is established.

(A one-sided test with a confidence level of 95% has the same power as a two- sided test with a confidence level of 90% , hence their appeal – if they can be justified)

There are some scenarios where you might argue that if the mean for the groups are not equal then one particular group of the two will likely be higher.

E.g. if men and women do not have equal earnings in an industry then it is probable than men earn more. Or if the prevalence of a medical condition is not the same for smokers and non-smokers then is it probable that the rate is higher for smokers.

However, there is an argument that one-sided tests should never (or at least rarely) be used. However probable we think it might be that if men and women haven't equal earnings then men earn more we cannot definitely say that it is impossible that it might be women who have the higher earnings and thus, it would be conduct a test that would never allow us to make such a finding.